

Erdős Problem #410

EP-410

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $\sigma_1(n) = \sigma(n)$, the sum of divisors function, and $\sigma_k(n) = \sigma(\sigma_{k-1}(n))$. Is it true that for all $n \geq 2$
 $\lim_{k \rightarrow \infty} \sigma_k(n)^{1/k} = \infty$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>